Examples of buffers

1. Acetic acid and sodium acetate mixture
2. Ammonium hydroxide and ammonium chloride mixture
3. Potassium dihydrogen phosphate and dipotassium hydrogen phosphate mixture.
4. Sodium carbonate and sodium bicarbonate mixture

Buffer action

Buffer solutions achieve their resistance to pH change because of the presence of an equilibrium between the acid HA and its conjugate base A^-.

$$HA \rightleftharpoons H^+ + A^-$$

When some strong acid is added to an equilibrium mixture of the weak acid and its conjugate base, the equilibrium is shifted to the left, in accordance with Le Chatelier's principle. Because of this, the hydrogen ion concentration increases by less than the amount expected for the quantity of strong acid added. Similarly, if strong alkali is added to the mixture the hydrogen ion concentration decreases by less than the amount expected for the quantity of alkali added.

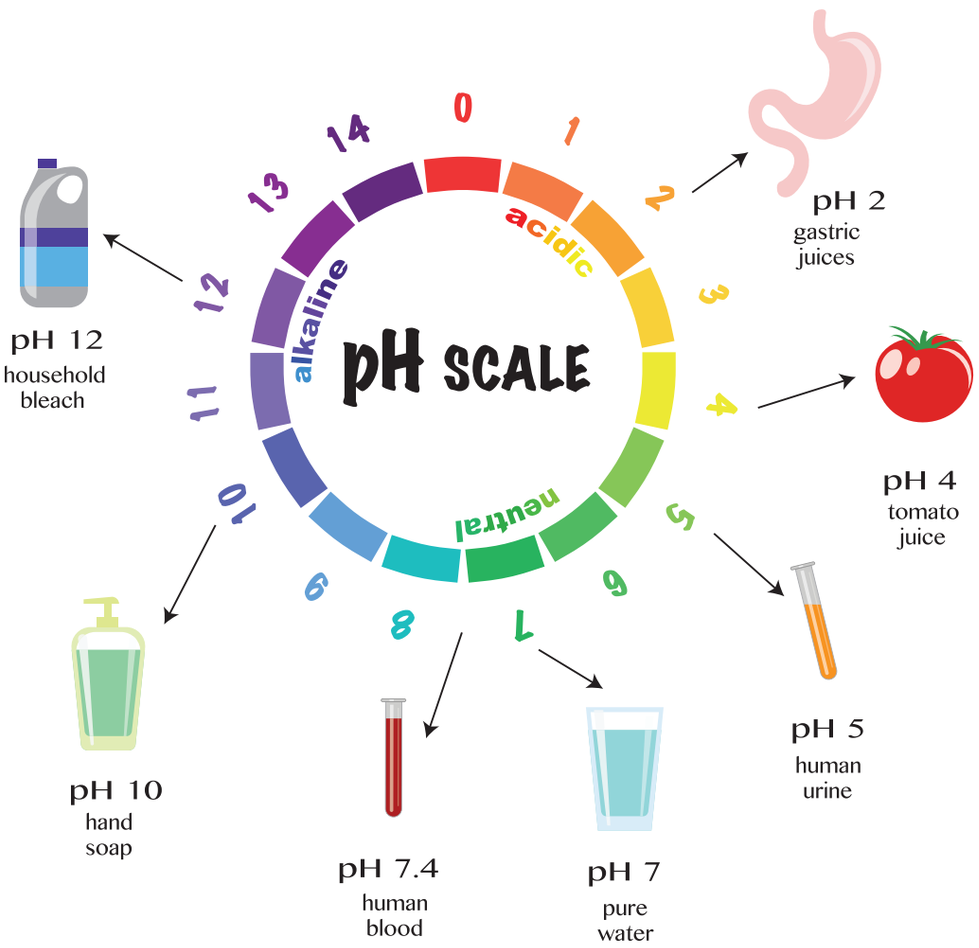


Figure 1.15 – pH paper and its colour change with respect to different samples.

Unit 1.indd 28 21-05-2018 15:07:39